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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

RESTAURANT ORDER PROCESSING AND BILLING SYSTEM

Submitted by

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DECLARATION

I hereby declare that the project entitled "**Parking Slot Allocation and Vehicle Entry Tracking System"** submitted in partial fulfillment of the requirements for the course **JAVA PROGRAMMING** is a **record of my original work** carried out under the guidance of **Mr.ARAVIND**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

Ihave acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

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CERTIFICATE

This is to certify that the project entitled **"ParkingSlotAllocationandVehicleEntryTrackingSystem"** is a **bona fide work** carried out by **SBNITHESHI"B"/2025BSCCOMPUTERSCIENCEKGCollegeof Arts & Science**, in partial fulfillment of the requirements for the course **JAVAPROGRAMMING**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date:10.04.2026

Submitted for the Viva-Voce Examination held on 10/04/2026

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ABSTRACT

The Parking Slot Allocation and Vehicle Entry Tracking System is developed to automate and efficiently manage vehicle parking in areas such as malls, offices, apartments, and public parking zones. Traditional parking systems rely heavily on manual processes, which often lead to confusion, time delays, improper slot utilization, and lack of security tracking.

This system provides a digital solution that allows automatic allocation of parking slots based on availability. It records vehicle entry and exit details, ensuring proper monitoring and management. The system helps reduce congestion, improve space utilization, and enhance security by maintaining accurate logs of all vehicles.

Additionally, the system supports features such as real-time slot availability, automated billing based on parking duration, and reporting tools for administrators. Overall, this system increases efficiency, reduces human error, and improves user experience in parking management.

Moreover, the system maintains a database of all vehicle entries and exits, which can be used for monitoring, reporting, and analysis. This enhances security by providing a clear record of all parked vehicles. The system is scalable and can be integrated with technologies such as RFID, sensors, or cameras for future enhancements. Overall, it improves operational efficiency, reduces human errors, and provides a seamless parking experience.

1.INTRODUCTION

The Parking Slot Allocation and Vehicle Entry Tracking System provides a smart solution by automating the entire parking process. It reduces dependency on manual work and ensures proper management of available space. The system uses a structured approach to allocate parking slots, monitor vehicle movements, and maintain accurate records.

By implementing this system, organizations can improve their parking management, reduce congestion, and enhance user satisfaction. It also provides better security by tracking every vehicle that enters and exits the premises.

1.1 Objectives

To automate parking slot allocation.
To track vehicle entry and exit details.
To reduce congestion and waiting time.
To improve parking space utilization.
To provide secure vehicle monitoring.
To generate billing based on parking duration.

1.2 Overview

The system integrates three major components:

1.Slot Allocation Module – Assigns available parking slots.

2.Vehicle Tracking Module – Records entry and exit time.

3.Billing & Reporting Module – Calculates parking charges and generates reports.

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1.3 Key Features

Real-Time Slot Availability – Displays available and occupied parking slots instantly to help drivers find space quickly.
Automatic Slot Allocation – Assigns parking slots automatically to vehicles based on availability and reduces manual effort.
Vehicle Entry Tracking – Records vehicle number and entry time to maintain accurate parking logs.
Vehicle Exit Tracking – Tracks exit time of vehicles and updates slot status for efficient management.
Automated Billing – Calculates parking charges based on duration and generates accurate invoices.
Multiple Payment Options – Supports cash, card, UPI, and digital payments for user convenience.
Database Management – Stores all vehicle and transaction details securely for future reference.
Report & Analytics – Provides reports on parking usage, revenue, and peak hours for better decision-making.
User-Friendly Interface – Simple and easy-to-use system for both users and administrators.
Security & Data Backup – Ensures safe storage of parking data with backup and monitoring features.

2.PROGRAM ANALYSIS

Programanalysisisacrucial phase in the development of the Parking Slot Allocation and Vehicle Entry TrackingSystem.Itinvolves understanding the existing problems in manual parking systems and identifying therequirementsforanimproved automated solution.

Thisphasehelpsindefining the system's functionality, scope, and limitations. By analyzing user needs and operationalchallenges,the system is designed to provide maximum efficiency and reliability.

2.1 Problem Definition

Manual parking systems are inefficient and often lead to several issues. Drivers spend a lot of time searching for available parking spaces, which leads to congestion and frustration. There is also a lack of proper monitoring, making it difficult to

track vehicles entering and leaving the parking area.
In addition, manual billing processes are prone to errors and lack transparency. Security is another concern, as there is no proper record of parked vehicles. These problems highlight the need for a computerized system that can automate parking management and improve efficiency.

2.2 Project Overview

The Parking Slot Allocation and Vehicle Entry Tracking System is designed as a comprehensive solution to manage parking operations effectively. The system automates the process of assigning parking slots, recording vehicle details, and calculating parking fees.

When a vehicle enters, the system assigns an available slot and records the entry time. Upon exit, it calculates the total time spent and generates a bill accordingly. The system also stores all records in a database for future reference.
This project aims to reduce manual work, improve accuracy, and provide a better experience for users. It also helps administrators manage parking facilities more efficiently.

2.3 Scopes and Limitations Scope

Applicable in malls and shopping centers.
Useful for office buildings.
Suitable for apartments and hospitals.
Supports multiple vehicles.
Can be expanded for smart parking systems.
Helps in managing large parking areas.

Limitations

Requires initial setup cost.

Needs electricity and system support.

Dependent on software and hardware.

Limited without upgrades.

Requires maintenance.

3.SYSTEM DESIGN

System designis a crucial phase in the development of the Parking Slot Allocation and Vehicle Entry Tracking System, wherethe overall structure and architecture of the application are planned before implementation. In this phase,thesystem is organized into multiple modules, each responsible for performing a specific function. This modularapproach helps in improving system efficiency, maintainability, and scalability.

The systemisdesigned as a user-friendly application that can be easily operated by parking staff or administrators.It follows a structured flow where the process begins with vehicle entry, continues with slot allocation,andends with billing and exit tracking. The design ensures that all operations are performed in a sequentialandlogical manner, reducing confusion and errors..

3.1 Modules

1. Vehicle Entry Module • Records

vehicle number • Captures entry

time

•Allocates parking slot

•Validates vehicle number format

•Prevents duplicate vehicle entry

•Sends entry details to database

2. Slot Allocation Module

•Checks available slots

•Assigns nearest/free slot

•Updates slot status

•Avoids assigning occupied slots

•Optimizes parking space usage

•Updates availability in real-time

3. Vehicle Exit Module

•Records exit time

•Frees the slot

•Verifies vehicle details before exit

•Updates database after exit

•Triggers billing process 4. Billing

Module •Calculates parking fee

•Based on time duration

•Generates receipt

4.METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the Hotel & Restaurant Billing System, a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

4.1. Problem Analysis

Parking areas face several challenges in managing vehicle movement, allocating slots, and tracking entry and exit details. Manual parking systems often result in:

Difficulty in finding available parking slots.

Time delays during vehicle entry and exit.

Lack of proper tracking of parked vehicles.

Inefficient use of parking space.

Errors in manual billing calculations.

This analysis highlights the need for an automated system that integrates slot allocation, vehicle tracking, and billing to improve efficiency, accuracy, and user convenience.

4.2. System Design

The system is designed with modular components:

Vehicle Entry Module – Records vehicle details and captures entry time.

Slot Allocation Module – Assigns available parking slots efficiently.

Vehicle Exit Module – Tracks exit time and frees the slot.

Billing Module – Calculates parking charges based on duration.

Database Module – Stores vehicle records and transaction details.

The design follows a structured approach with a centralized database to manage parking data and ensure smooth system operation

4.3. Program Development

- Developed using programming languages such as Java/Python/C# with a database like MySQL/SQLite.
- User interface is created using GUI frameworks or web-based technologies for easy interaction.
- Modules are developed separately and integrated to form a complete parking management system.
- Focus is given to modularity, scalability, and maintainability for future enhancements.

4.4. Testing and Validation

Testing ensures that the system functions correctly. Various test cases are executed to verify slot allocation, vehicle tracking, and billing accuracy.

4.5. Result and Documentation.

The system successfully automates parking management operations.

Reduces manual work and human errors.

Improves efficiency in vehicle entry and exit handling.

Ensures proper allocation of parking slots.

Provides real-time tracking of parked vehicles.

Minimizes waiting time for users.

Enhances utilization of available parking space.

Generates accurate billing based on parking duration.

Maintains transparency in the billing process.

Stores all data securely in a centralized database.

Allows easy retrieval of parking records.

Improves overall user experience and convenience.

4.6. Algorithm

Algorithm for the **RESTAURANT ORDER PROCESSING AND BILLING SYSTEM**

Step 1: Start

Step 2:Check available slots

Step 3:If slot available → Assign slot

Step 4:Record vehicle entry time

Step 5: On exit → Record exit time

Step 6: Calculate duration

Step 7: Generate bill

Step 8: Free slot

Step 9: End

4.7. Flowchart

Start

↓

Check Slot Availability

↓

Assign Slot

↓

Record Entry Time

↓

Vehicle Exit

↓

Calculate Time

↓

Generate Bill

↓

Free Slot

↓

End

5.SOURCE CODE

```
import java.util.*;

class ParkingSystem {
   static int totalSlots = 5;
   static String[] slots = new String[totalSlots];
   static Scanner sc = new Scanner(System.in);

   public static void main(String[] args) {
      int choice;

      do {
         System.out.println("\n--- Parking
System ---");
         System.out.println("1. Park Vehicle");
         System.out.println("2. Remove
Vehicle");
         System.out.println("3. View Slots");
         System.out.println("4. Exit");
         System.out.print("Enter choice: ");
         choice = sc.nextInt();

         switch (choice) {
            case 1:
               parkVehicle();
               break;
            case 2:
               removeVehicle();
               break;
            case 3:
               viewSlots();
               break;
            case 4:
               System.out.println("Exiting...");
               break;
            default:
               System.out.println("Invalid choice!");
         }
      } while (choice != 4);
   }

   // Function to park vehicle
   static void parkVehicle() {
      System.out.print("Enter vehicle number:
");
      String vehicle = sc.next();

      for (int i = 0; i < totalSlots; i++) {
         if (slots[i] == null) {
            slots[i] = vehicle;


}
```

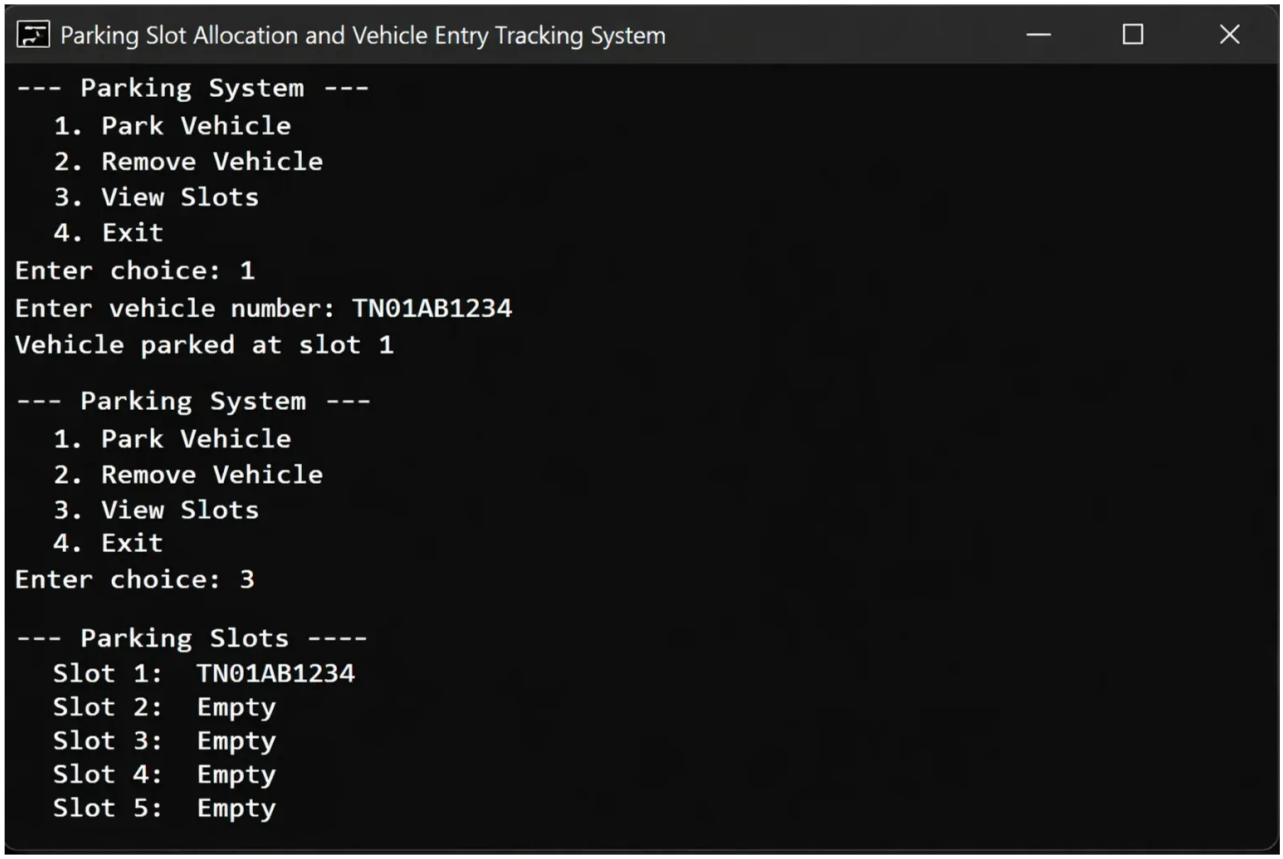
```
System.out.println("Vehicle parked at
slot " + (i + 1));
            return;
          }
      }
      System.out.println("Parking Full!");
   }

   // Function to remove vehicle
   static void removeVehicle() {
      System.out.print("Enter vehicle
number to remove: ");
      String vehicle = sc.next();

      for (int i = 0; i < totalSlots; i++) {
         if (vehicle.equals(slots[i])) {
            slots[i] = null;
            System.out.println("Vehicle
removed from slot " + (i + 1));
            return;
         }
      }
      System.out.println("Vehicle not
found!");
   }

   // Function to view slots
   static void viewSlots() {
      System.out.println("\n--- Parking
Slots ---");
      for (int i = 0; i < totalSlots; i++) {
         if (slots[i] == null) {
            System.out.println("Slot " + (i + 1)
+ ": Empty");
         } else {
            System.out.println("Slot " + (i + 1)
+ ": " + slots[i]);
         }
      }
   }
}
```

6.INPUT AND OUTPUT SCREENS



7.DISCUSSION & CONCLUSION

7.1 Discussion

The Parking Slot Allocation and Vehicle Entry Tracking System was developed to overcome the limitations of traditional manual parking systems. During the implementation, the system proved to be effective in managing parking operations in a structured and efficient manner. The integration of modules such as vehicle entry, slot allocation, vehicle exit, billing, and database management ensures smooth workflow and coordination.

Testing results showed that the system reduces human errors, improves speed in parking operations, and ensures accurate tracking of vehicles. The automatic slot allocation feature helps in better space utilization, while real-time updates prevent conflicts such as assigning the same slot to multiple vehicles. The billing module generates accurate charges based on parking duration, improving transparency.

7.2 Conclusion

The Parking Slot Allocation and Vehicle Entry Tracking System is a reliable and efficient solution for modern parking challenges. It successfully automates parking operations such as slot allocation, vehicle tracking, and billing, thereby reducing manual work and minimizing errors. The system ensures optimal utilization of parking space and reduces time spent by users in searching for available slots.

By maintaining accurate records of vehicle entry and exit, the system enhances security and provides better monitoring of parking activities. The automated billing process ensures transparency and accuracy, improving user trust and satisfaction. Additionally, the system's modular design allows easy maintenance and future upgrades, making it adaptable to different environments.

8.REFERENCES

The following resources were used during the development of this project:

Books

Software Engineering: A Practitioner's Approach

Software Engineering

Websites

W3Schools – https://www.w3schools.com

Oracle – https://docs.oracle.com

GeeksforGeeks – https://www.geeksforgeeks.org